

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial 6 - Continuous fermentation with circulation and concentration – Solutions

Question 1

For the purpose of extracellular protein production, a factory grows yeast in continuous fermentation with circulation and concentration. The fermenter volume is 625 liter and the fresh feed flow rate is $0.5 \frac{m^3}{hr}$. The circulation factor of the medium into the fermenter is 0.3 and the concentration coefficient of the continuous centrifugation is 1.8.

The substrate concentration in the fresh feed is 2% and the product yield on this substrate is $0.27 \frac{gP}{gS}$. The maximal growth rate of the yeast in the given medium is 1.1 hr^{-1} and Monod constant is $8.69 \frac{g}{lit}$.

Due to economic reasons, the factory manager decided to increase the production rate by factor of 2. To do so, three options were considered:

Engineer (1) suggest to increase the circulation factor.

Engineer (2) suggest to increase the concentration coefficient of the centrifugation.

Engineer (3) suggest to increase the fresh feed flow rate.

- A. What is the total product production rate ($\frac{gP}{hr}$) of the given fermentation, before the change is executed?
- B. Theoretically, without performing any calculations, which of the engineer suggestions is the most right? Detail and explain by using parametric equations if needed, for each of the suggestions given.

C. Choose one of the engineers' suggestions that is the most relevant to multiply the total product production rate in this fermentation. Calculate what is the change that should be done in practice (use the data given for this specific fermentation).

Solution:

Given:

Fermentation parameters

$$V = 625 \text{ liter}$$

$$F = 0.5 \frac{m^3}{hr} = 500 \frac{lit}{hr}$$

$$w = 0.3$$

$$C = 1.8$$

$$S_0 = 2\% = 20 \frac{gS}{lit}$$

$$Y_{\frac{P}{S}} = 0.27 \frac{gP}{gS}$$

$$\mu_{max} = 1.1 \frac{1}{hr}$$

$$K_s = 8.69 \frac{g}{lit}$$

Improved fermentation

$$G_{p_new} = 2G_p$$

$$(1) w \uparrow$$

$$(2) C \uparrow$$

$$(3) F \uparrow$$

A. $G_p = ?$

The production rate will be calculated using: $G_p = F \cdot P_F$.

The flow rate, F , is given in the question. We need to find P_F using the different balances that we learned.

First, biomass balance:

$$\frac{dx}{dt} = \frac{F}{V} \cdot X_0 + \frac{w \cdot F}{V} \cdot C \cdot X_F + \mu \cdot X_F - \alpha \cdot X_F - \frac{(1 + w) \cdot F}{V} \cdot X_F$$

Assumptions:

- Steady state $\frac{dx}{dt} = 0$
- Sterile feed $X_0 = 0$
- $\alpha \ll \mu$

We get the following: $\mu = D \cdot [1 + w \cdot (1 - C)]$

Second, substrate balance:

$$\frac{ds}{dt} = \frac{F}{V} \cdot S_0 + \frac{w \cdot F}{V} \cdot S_F - \frac{\mu \cdot X_F}{Y_{x/s}} - \frac{q_p \cdot X_F}{Y_{p/s}} - m_s \cdot X_F - \frac{(1 + w) \cdot F}{V} \cdot S_F$$

Assumptions:

- Steady state $\frac{ds}{dt} = 0$
- Negligible $\frac{q_p \cdot X_F}{Y_{p/s}}, m_s \cdot X_F$

We get the following: $\frac{\mu \cdot X_F}{Y_{x/s}} = D \cdot (S_0 - S_F)$

Place μ as was defined above to receive: $X_F = \frac{Y_{x/s}(S_0 - S_F)}{[1 + w \cdot (1 - C)]}$

Third, product balance:

$$\frac{dp}{dt} = \frac{F}{V} \cdot P_0 + \frac{w \cdot F}{V} \cdot P_F + q_p \cdot X_F - K \cdot P_F - \frac{(1 + w) \cdot F}{V} \cdot P_F$$

Assumptions:

- Steady state $\frac{dp}{dt} = 0$
- $P_0 = 0$
- Negligible $K P_F$

Place μ and X_F to receive: $P_F = Y_{p_s} \cdot (S_0 - S_F)$.

Now we have an equation that can help us calculate P_F .

Y_{p_s} and S_0 are given in the question so we only need to find S_F . We can do it by using

Monod equation:

$$S_F = \left(\frac{\mu \cdot K_s}{\mu_{max} - \mu} \right)$$

We have all the parameters except μ , that can be calculated according to the equation that was developed from the biomass balance:

$$\mu = D \cdot [1 + w \cdot (1 - C)] = \frac{F}{V} \cdot [1 + w \cdot (1 - C)]$$

$$\mu = \frac{500 \frac{lit}{hr}}{625 \frac{lit}} \cdot [1 + 0.3 \cdot (1 - 1.8)] = 0.608 \frac{1}{hr}$$

Now we can place it in the equation above to find S_F :

$$S_F = \left(\frac{\mu \cdot K_s}{\mu_{max} - \mu} \right) = \left(\frac{0.608 \frac{1}{hr} \cdot 8.69 \frac{g}{lit}}{(1.1 - 0.608) \frac{1}{hr}} \right) = 10.74 \frac{gS}{lit}$$

Place it back to calculate P_F :

$$P_F = Y_{p_s} \cdot (S_0 - S_F) = 0.27 \frac{gP}{gS} \cdot (20 - 10.74) \frac{gS}{lit} = 2.5 \frac{gP}{lit}$$

Finally, place numbers to find the total production rate:

$$G_p = F \cdot P_F = 500 \frac{lit}{hr} \cdot 2.5 \frac{gP}{lit}$$

$$G_p = 1250 \frac{gP}{hr}$$

B.

We first need to understand how does each of the suggested changes affect the size of F or P_F (since $G_p = F \cdot P_F$).

Engineer (1) suggests to increase the circulation factor w .

Let's look on the expression that was developed to describe the growth rate:

$$\mu = D[1 + w \cdot (1 - C)]$$

w is a fraction ($w < 1$), so if w increase it will cause μ to decrease.

If μ decrease then also S_F decrease, as rises from Monod equation:

$$S_F = \left(\frac{\mu \cdot K_s}{\mu_{max} - \mu} \right)$$

The correlation between S_F and P_F is described by the following equation:

$$P_F = Y_{\frac{p}{s}} \cdot (S_0 - S_F)$$

$Y_{\frac{p}{s}}$ and S_0 are constant, so if S_F decrease it will increase the product concentration (P_F) in the fermenter, result in **higher production rate (G_p)**.

Engineer (2) suggests to increase the concentration factor C .

In a similar way, the biomass equation reveals that when C increase it lowers the growth rate (μ), which in turn decrease S_F that lead to **increase** in P_F and G_p .

Engineer (3) suggests to increase the flow rate F .

Pay attention! It is not enough to only look on $G_p = F \cdot P_F$.

F is also involve on the growth rate equation: $\mu = \frac{F}{V} \cdot [1 + w \cdot (1 - C)]$

So if F increase it will increase μ , that in turn increase S_F but decrease P_F .

Additionally, increasing F is limited until rinsing process occur. Moreover, increasing F and S_F causing wastage of substrate (so it's not cost-effective).

To summarize, we can't know what will happen to G_p .

C.

As was shown in section B, the only relevant suggestions are of engineer (1) and (2).

First, we need to make a quick test that will tell us which of the suggestions can bring to more significant change in the production. Such quick test can be done by placing numbers (all parameters are given in the question) in the growth rate equation:

$$\mu = \frac{F}{V} \cdot [1 + w \cdot (1 - C)] \text{ and compare to the original size } \mu = 0.608 \frac{1}{hr}.$$

According to engineer (1) suggestion: $w' = 2w$

$$\mu = \frac{500 \frac{lit}{hr}}{625 lit} \cdot [1 + 0.6 \cdot (1 - 1.8)] = 0.416 \frac{1}{hr}$$

According to engineer (2) suggestion: $C' = 2C$

$$\mu = \frac{500 \frac{lit}{hr}}{625 lit} \cdot [1 + 0.3 \cdot (1 - 3.6)] = 0.176 \frac{1}{hr}$$

Engineer (2) suggestion reduces μ more drastically, so we will choose it. Solution steps:

- 1) We aim to multiply G_p by 2, so we will find the new size of G_{pnew} .
- 2) Find the new product concentration (P_{Fnew}).
- 3) Find the new growth rate (μ).
- 4) Find the new C .

$$G_{pnew} = 2 \cdot 1250 \frac{gP}{hr} = 2500 \frac{gP}{hr}$$

The product concentration is directly related to the production rate so:

$$G_{pnew} = F \cdot P_{Fnew}$$

$$P_{Fnew} = \frac{2500 \frac{gP}{hr}}{500 \frac{lit}{hr}} = 5 \frac{gP}{lit}$$

We can use Monod equation to find μ :

$$\mu = \mu_{max} \cdot \left(\frac{S}{K_s + S} \right)$$

We have μ_{max} and K_s given in the question, so we only need to find S_F .

S_F can be calculated using the product balance development:

$$P_F = Y_{\frac{P}{S}} \cdot (S_0 - S_F)$$

$$S_F = S_0 - \frac{P_F}{Y_{\frac{P}{S}}} = 20 \frac{gS}{lit} - \frac{5 \frac{gP}{lit}}{0.27 \frac{gP}{gS}} = 1.48 \frac{gS}{lit}$$

Back to Monod:

$$\mu = 1.1 \frac{1}{hr} \cdot \left(\frac{1.48 \frac{gS}{lit}}{(8.69 + 1.48) \frac{gS}{lit}} \right) = 0.16 \frac{1}{hr}$$

Lastly, we will calculate the new C :

$$\mu = \frac{F}{V} \cdot [1 + w \cdot (1 - C)]$$

$$0.16 \frac{1}{hr} = \frac{500 \frac{lit}{hr}}{625 lit} \cdot [1 + 0.3 \cdot (1 - C)]$$

$$C = 3.67$$

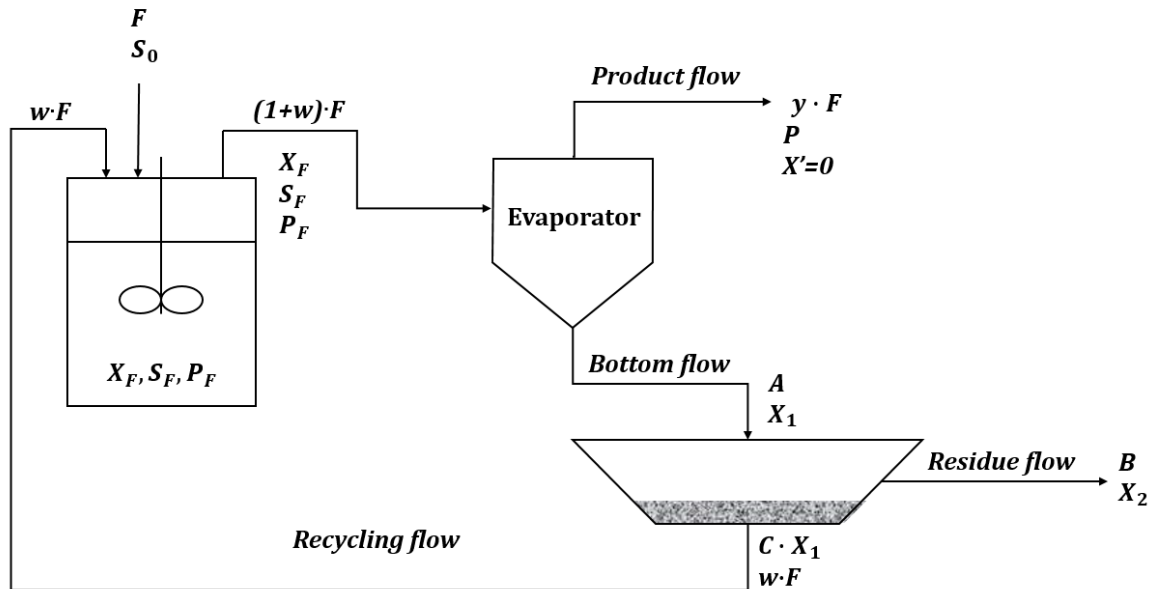
* Note – the suggestion of engineer (1) is not possible since:

$$0.16 \frac{1}{hr} = \frac{500 \frac{lit}{hr}}{625 lit} \cdot [1 + w(1 - 1.8)] \rightarrow w = 1$$

But as usually the working range of w is $0 < w < 1$, if the required w is $w = 1$ – it is not the preferred option.

Question 2

A factory performs continuous fermentation with distillation, circulation and concentration. The product is a volatile compound with a special aroma, that can be produced by a yeast that secret it to the medium. The system is described below:



The fermenter is being continuously supplied with a fresh medium that contains substrate only. The liquid that exit the fermenter is being transferred to a continuous evaporator, from which 2 flows are exiting: product flow with a flow rate of $y \cdot F$, that contains only product and water, and bottom flow with a flow rate of A and yeast concentration of X_1 . The bottom flow runs through a continuous centrifuge, from which another 2 flows are exiting: residue flow with a flow rate of B and biomass concentration of X_2 , and circulation flow with a flow rate of $w \cdot F$ and biomass concentration of $C \cdot X_1$.

Additional fermentation data: $C = 2$, $w = 0.4$, $y = 0.3$, $D = 0.2625 \text{ hr}^{-1}$, $X_2 = 0.857 \frac{\text{g}}{\text{lit}}$.

All flow rates (F, A, B) are given in $\frac{\text{lit}}{\text{hr}}$.

All the standard assumptions that were taken throughout the equations development are valid (except for assumptions that are contradicted in this question).

What is the growth rate in the fermenter?

Solution:

$$\mu = ?$$

Given:

$$C = 2$$

$$w = 0.4$$

$$y = 0.3$$

$$D = 0.2625 \frac{1}{hr}$$

$$X_2 = 0.857 \frac{g}{lit}$$

We will start with the biomass balance which link μ to other parameters of the fermentation:

$$\frac{dx}{dt} = \frac{F}{V} \cdot X_0 + \frac{w \cdot F}{V} \cdot CX_1 + \mu \cdot X_F - \alpha \cdot X_F - \frac{(1+w) \cdot F}{V} \cdot X_F$$

Assumptions:

- Steady state $\frac{dx}{dt} = 0$
- Sterile feed $X_0 = 0$
- $\alpha \ll \mu$
- $D = \frac{F}{V}$

$$\text{We get the following: } w \cdot D \cdot CX_1 + \mu \cdot X_F - D \cdot (1+w) \cdot X_F = 0$$

And the final equation is:

$$[1] \mu = \frac{D \cdot [(1+w) \cdot X_F - w \cdot C \cdot X_1]}{X_F}$$

We know the size of D , w and C , so we only need to find X_F and X_1 .

The data given in the question is related to X , so we should work with balances on X (and not S or P).

We start by writing balance on X in the evaporator and concentrator (centrifuge):

Evaporator:

$$\frac{dx}{dt} = (1 + w) \cdot F \cdot X_F - y \cdot F \cdot X' - A \cdot X_1$$

We assume that the system is in steady state and it is given that $X' = 0$, so we get:

$$[2] (1 + w) \cdot F \cdot X_F - A \cdot X_1 = 0$$

We don't have information on F so we need to express it via other parameters that are known or can be found.

Mass balance on the flow rates:

$$(1 + w) \cdot F = y \cdot F + A \rightarrow F = \frac{A}{1+w-y}$$

Place F in equation [2] to get an expression for X_F :

$$[3] X_F = X_1 \cdot \frac{1+w-y}{1+w}$$

We need to find X_1 . Let's continue to balances on the centrifuge.

Centrifuge:

Balance on the flow rates:

$$A = B + w \cdot F \rightarrow B = A - w \cdot F$$

Biomass balance:

$$\frac{dx}{dt} = A \cdot X_1 - B \cdot X_2 - w \cdot F \cdot C \cdot X_1 = 0$$

Place B :

$$A \cdot X_1 - (A - w \cdot F) \cdot X_2 - w \cdot F \cdot C \cdot X_1 = 0$$

$$X_1 \cdot (A - w \cdot F \cdot C) = X_2 \cdot (A - w \cdot F)$$

Place F as was found above:

$$X_1 \cdot \left(A - \frac{w \cdot C \cdot A}{1 + w - y} \right) = X_2 \cdot \left(A - \frac{w \cdot A}{1 + w - y} \right)$$

Divide by A and multiply by $(1 + w - y)$ to obtain the following equation:

$$X_1 \cdot (1 + w - y - w \cdot C) = X_2 \cdot (1 + \cancel{w} - y - \cancel{w})$$

$$X_1 = X_2 \cdot \left(\frac{1 - y}{1 - y + w \cdot (1 - C)} \right)$$

We have all numbers to find X_1 :

$$X_1 = 0.857 \frac{g}{lit} \cdot \left(\frac{1 - 0.3}{1 - 0.3 + 0.4 \cdot (1 - 2)} \right) = 2 \frac{g}{lit}$$

* **Note!** That $X_1 > X_2$ as expected from the described system.

Back to equation [3] to find X_F :

$$X_F = X_1 \cdot \left(\frac{1 + w - y}{1 + w} \right) = 2 \frac{g}{lit} \cdot \left(\frac{1 + 0.4 - 0.3}{1 + 0.4} \right) = 1.57 \frac{g}{lit}$$

* **Note!** It makes sense that $X_1 > X_F$ because during evaporation we are losing water, so the volume reduces and the concentration rises.

Now we can finally go back to equation [1] to find μ :

$$\mu = \frac{0.2625 \frac{1}{hr} \cdot [(1 + 0.4) \cdot 1.57 \frac{g}{lit} - 0.4 \cdot 2 \cdot 2 \frac{g}{lit}]}{1.57 \frac{g}{lit}}$$

$$\mu = 0.1 \frac{1}{hr}$$

Question 3

You were accepted to a new startup company, and as the leading biotechnology engineer you were asked to plan a fermentation in a new 1000 liter fermenter. The fermentation aims to produce a special hormone, wanted by the pharmaceutical industry. The hormone can be produced by a mutant yeast that was developed in this company.

The hormone product is extracellular and the production yield is $0.25 \frac{g \text{ hormone}}{g \text{ yeast}}$. The carbon source in the fermentation is glucose and the yeast growth yield is $0.5 \frac{g \text{ yeast}}{g \text{ glucose}}$. The yeast maximal growth rate in the given medium is 0.48 hr^{-1} and Monod constant is $0.5 \frac{g}{lit}$.

You have to decide in which fermentation method should be chosen to produce the product in minimal cost (meaning, the production cost of 1 kg product should be as low as possible). It is given that 1 working hour in this factory cost 400\$ (this cost includes labor and other constants costs such as electricity and similar) and that 1 Kg of substrate costs 100\$ (the cost is based on the carbon source weight, but it includes also the other medium ingredients).

Since the company is new, it still doesn't have all the required instruments, so the feed medium for the fermentation is not prepared in the factory but bought as prepared medium with sugar concentration of 3%. Additionally, there are some other engineering limitations that will be detailed later.

You can choose one of following fermentation options:

- I. **Batch fermentation:** in this case, the initial yeast concentration in the fermenter is $0.1 \frac{g}{lit}$ and the growth rate, in average, is 95% of the maximal growth rate. Upon yeast inoculation, it stays in lag stage for 1 hour, and at the end of the fermentation it takes 3 hours for discharge, cleaning, filling and sterilization, before the fermenter is ready for the next fermentation.
- II. **Continuous fermentation:** in this case, the maximal flow rate of the fresh medium into the fermenter is $400 \frac{lit}{hr}$ (limitation of the available pump...).
- III. **Continuous fermentation with circulation and concentration:** in this case, in addition to the maximal flow rate of the fresh medium that was mentioned in option II, it is also given that the possible coefficients for circulation and concentration are 0.3 and 2.5, respectively.

Which of the abovementioned fermentation methods you would choose, in order to ensure a minimal production cost of 1 kg hormone?

Solution:

Given:

<u>General data</u>	<u>Batch</u>	<u>Continuous</u>	<u>with circulation and concentration</u>
$V = 1000 \text{ liter}$	$X_0 = 0.1 \frac{gX}{lit}$	$F_{max} = 400 \frac{lit}{hr}$	$w = 0.3$
$Y_{\frac{P}{X}} = 0.25 \frac{gP}{gX}$	$\mu = 0.95 \cdot \mu_{max}$		$C = 2.5$
$Y_{\frac{X}{S}} = 0.5 \frac{gX}{gS}$	$= 0.456 \frac{1}{hr}$		
$\mu_{max} = 0.48 \frac{1}{hr}$	$t_{lag} = 1 \text{ hr}$		
$K_S = 0.5 \frac{g}{lit}$	$t_{maintenance} = 3 \text{ hr}$		
$S_0 = 3\% = 30 \frac{gS}{lit}$			
$cost_{work_{hr}} = 400 \frac{\$}{hr}$			
$cost_{kgS} = 100 \frac{\$}{kgS}$			

min_cost_{kg_p} = ?

We need to find how much it costs to produce 1 kg product in each of the proposed systems in order to know what is the minimal cost.

Remember that in order to calculate the total cost of 1 kg product production in each fermentation method we need to find the amount of working hours (growth time dependent) and the amount of substrate required.

[I] Batch fermentation:

First, we need to find out how much product is being produced in one fermentation cycle.

From a look on the given data, we can use the yields to calculate P_F :

$$Y_{\frac{P}{S}} = Y_{\frac{P}{X}} \cdot Y_{\frac{X}{S}} = \frac{\Delta P}{\Delta S} = \frac{P_F - \cancel{P_0}}{S_0 - \cancel{S_F}} = \frac{P_F}{S_0}$$

$$P_F = S_0 \cdot \left(Y_{\frac{P}{X}} \cdot Y_{\frac{X}{S}} \right) = 30 \frac{gS}{lit} \cdot \left(0.25 \frac{gP}{gX} \cdot 0.5 \frac{gX}{gS} \right) = 3.75 \frac{gP}{lit}$$

To know the total product weight, we should multiply by the fermenter volume:

$$W_{P_F} = V \cdot P_F = 1000 \text{ lit} \cdot 3.75 \frac{gP}{lit} \cdot \frac{1 \text{ kgP}}{1000 \text{ gP}} = 3.75 \text{ kgP}$$

Second, we need to calculate the total period of the fermentation:

$$t_{tot} = t_{lag} + t_{log} + t_{maintenance}$$

We need to calculate t_{log} :

$$\ln\left(\frac{X_F}{X_0}\right) = \mu \cdot t_{log}$$

We have information on X_0 and μ , so we only need to find X_F . We will use the product yield:

$$Y_{\frac{P}{X}} = \frac{P_F - \cancel{P_0}}{X_F - X_0}$$

We previously found P_F , and $Y_{\frac{P}{X}}$ and X_0 are given in the question:

$$X_F = \frac{P_F}{Y_{\frac{P}{X}}} + X_0 = \frac{3.75 \frac{gP}{lit}}{0.25 \frac{gP}{gX}} + 0.1 \frac{gX}{lit} = 15.1 \frac{gX}{lit}$$

Back to the growth equation:

$$t_{log} = \frac{\ln\left(\frac{X_F}{X_0}\right)}{\mu} = \frac{\ln\left(\frac{15.1 \frac{gX}{lit}}{0.1 \frac{gX}{lit}}\right)}{0.456 \frac{1}{hr}} = 11 \text{ hr}$$

Therefore, the total period of 1 cycle of fermentation is:

$$t_{tot} = 1 \text{ hr} + 11 \text{ hr} + 3 \text{ hr} = 15 \text{ hr}$$

Third, we need to calculate the amount of substrate required for this fermentation:

$$W_{S_0} = V \cdot S_0 = 1000 \text{ lit} \cdot 30 \frac{\text{gS}}{\text{lit}} \cdot \frac{1 \text{ kgP}}{1000 \text{ gP}} = 30 \text{ kgS}$$

Last, we have all we need to calculate the total cost of this fermentation ($Cost_f$):

$$Cost_f = t_{tot} \cdot Cost_{work_{hr}} + W_{S_0} \cdot Cost_{kgS} = 15 \text{ hr} \cdot 400 \frac{\$}{\text{hr}} + 30 \text{ kgS} \cdot 100 \frac{\$}{\text{kgS}} = 9000 \$$$

This cost is per 1 fermentation but we want to know what is the cost of 1 kg product ($Cost_{kgP}$). We previously found how much product is being produced per fermentation (3.75 kgP), so we can use it to normalize the total cost to be per 1 kg product:

$$Cost_{kgP} = \frac{9000 \$}{3.75 \text{ kgP}} = 2400 \frac{\$}{\text{kgP}}$$

[II] Continuous fermentation:

This is a case of standard continuous fermentation so we can use the standard balances equations and assumptions.

We will check how much time and substrate it takes to produce 1 kg product in this fermentation.

$$\text{Product flow rate} - G_P = F \cdot P_F$$

We need to find both F and P_F .

From biomass balance we get that: $\mu = D = \frac{F}{V}$

$$\mu = \frac{400 \frac{\text{lit}}{\text{hr}}}{1000 \text{ lit}} = 0.4 \frac{1}{\text{hr}}$$

We must make sure that the current flow rate of the pump doesn't lead us to rinsing, meaning: the given F_{max} must be lower than F_c

$$F_c = D_c \cdot V = \mu_{max} \cdot V = 0.48 \frac{1}{hr} \cdot 1000 \text{ lit} = 480 \frac{\text{lit}}{hr} \rightarrow F_c > F_{max} \rightarrow \text{no rinsing}$$

From product balance we get that: $P_F = Y_P \cdot (S_0 - S_F)$

We have $Y_P (= Y_X \cdot Y_P)$ and S_0 . We can find S_F from Monod equation:

$$S_F = \frac{\mu \cdot K_s}{\mu_{max} - \mu} = \frac{0.4 \frac{1}{hr} \cdot 0.5 \frac{g}{lit}}{(0.48 - 0.4) \frac{1}{hr}} = 2.5 \frac{g}{lit}$$

Back to calculate P_F :

$$P_F = Y_P \cdot (S_0 - S_F) = 0.25 \frac{gP}{gX} \cdot 0.5 \frac{gX}{gS} \cdot (30 - 2.5) \frac{gS}{lit} = 3.43 \frac{gP}{lit}$$

Therefore, the production flow rate is:

$$G_P = 400 \frac{\text{lit}}{hr} \cdot 3.43 \frac{gP}{lit} \cdot \frac{1 \text{ kgP}}{1000 gP} = 1.372 \frac{\text{kgP}}{hr}$$

Now we can calculate how much time it takes to produce 1 kg product:

$$t = \frac{1 \text{ kgP}}{1.372 \frac{\text{kgP}}{hr}} = 0.73 \text{ hr}$$

We still need the amount of substrate required:

$$G_{S_0} = 400 \frac{\text{lit}}{hr} \cdot 30 \frac{gS}{lit} \cdot \frac{1 \text{ kgP}}{1000 gP} = 12 \frac{\text{kgP}}{hr}$$

$$W_{S_0} = t \cdot G_{S_0} = 0.73 \text{ hr} \cdot 12 \frac{\text{kgP}}{hr} = 8.76 \text{ kgS}$$

Last, we have all we need to calculate the total cost of this fermentation ($Cost_f$):

$$Cost_{kgP} = t_{tot} \cdot Cost_{work_{hr}} + W_{S_0} \cdot Cost_{kgS} = 0.73 \text{ hr} \cdot 400 \frac{\$}{hr} + 8.76 \text{ kgS} \cdot 100 \frac{\$}{\text{kgS}}$$

$$\mathbf{Cost_{kgP} = 1168 \$}$$

[III] Continuous fermentation with circulation and concentration:

Similarly, we can use the standard equations and progress as we did for option (II).

$$\mu = [1 + w \cdot (1 - C)] \cdot \frac{F}{V} = [1 + 0.3 \cdot (1 - 2.5)] \cdot \frac{400 \frac{\text{lit}}{\text{hr}}}{1000 \text{ lit}} = 0.22 \frac{1}{\text{hr}}$$

$$S_F = \frac{\mu \cdot K_s}{\mu_{\max} - \mu} = \frac{0.22 \frac{1}{\text{hr}} \cdot 0.5 \frac{\text{g}}{\text{lit}}}{(0.48 - 0.22) \frac{1}{\text{hr}}} = 0.423 \frac{\text{g}}{\text{lit}}$$

$$P_F = Y_{\frac{P}{S}} \cdot (S_0 - S_F) = 0.25 \frac{\text{gP}}{\text{gX}} \cdot 0.5 \frac{\text{gX}}{\text{gS}} \cdot (30 - 0.423) \frac{\text{gS}}{\text{lit}} = 3.7 \frac{\text{gP}}{\text{lit}}$$

$$G_P = 400 \frac{\text{lit}}{\text{hr}} \cdot 3.7 \frac{\text{gP}}{\text{lit}} \cdot \frac{1 \text{ kgP}}{1000 \text{ gP}} = 1.48 \frac{\text{kgP}}{\text{hr}}$$

$$t = \frac{1 \text{ kgP}}{1.48 \frac{\text{kgP}}{\text{hr}}} = 0.676 \text{ hr}$$

The size of G_{S_0} does not change, so the amount of substrate consumed during the fermentation period is:

$$W_{S_0} = t \cdot G_{S_0} = 0.676 \text{ hr} \cdot 12 \frac{\text{kgP}}{\text{hr}} = 8.1 \text{ kgS}$$

$$\text{Cost}_{\text{kgP}} = t_{\text{tot}} \cdot \text{Cost}_{\text{work}_{\text{hr}}} + W_{S_0} \cdot \text{Cost}_{\text{kgS}} = 0.676 \text{ hr} \cdot 400 \frac{\$}{\text{hr}} + 8.1 \text{ kgS} \cdot 100 \frac{\$}{\text{kgS}}$$

$$\text{Cost}_{\text{kgP}} = 1081 \$$$

To conclude, the minimal production cost will be achieved from method (III), continuous fermentation with circulation and concentration.